

NUMBERS AND ARITHMETIC OPERATIONS

1.1 NUMBERS

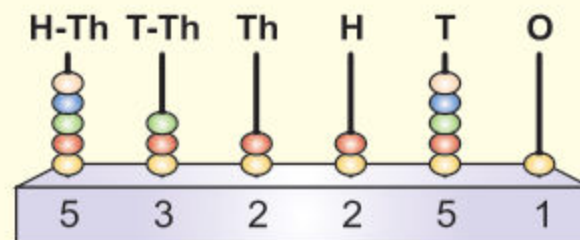
In class III, we have learnt counting objects and writing of numbers up to hundred thousand on the basis of place value of numbers.

Let us revise

Example 1: Number “Five hundred thirty two thousand two hundred fifty one” is written in figures as:

H-Th	T-Th	Th	H	T	O
5	3	2	2	5	1

In abacus, it is represented.



$$500,000 + 30,000 + 2,000 + 200 + 50 + 1 = \boxed{532,251}$$

EXERCISE 1.1

1 Write the following numbers in figures.

- (i) Four thousand six hundred and ninety two.
- (ii) Sixty nine thousand four hundred and seven.
- (iii) Four hundred fifty six thousand and nine hundred twenty six.
- (iv) Seven hundred thousand.
- (v) Nine hundred two thousand and forty two.
- (vi) Seven hundred twenty nine thousand and six.

2 Write the following numbers in words.

- (i) 2,561
- (ii) 34,561
- (iii) 56,081
- (iv) 92,000
- (v) 245,612
- (vi) 349,650

Unit 1 NUMBERS AND ARITHMETIC OPERATIONS (Numbers)

Identify place values of digits up to one hundred million

Place value chart for one million

The smallest seven digit number is called one million,

i.e. **1,000,000**

Place value chart for **1,000,000**

Millions	Hundred Thousands	Ten Thousands	Thousands	Hundreds	Tens	Ones
M	H-Th	T-Th	Th	H	T	O
1	0	0	0	0	0	0

Place value chart for ten million

Greatest seven digit number is **9,999,999**

“Nine million nine hundred and ninety nine thousand nine hundred and ninety nine”

When we add 1 to 9,999,999, we get **10,000,000**. Thus the number after 9,999,999 is 10,000,000 Read as “Ten Million”
i.e. $9,999,999 + 1 = 10,000,000$

Place Value Chart for **10,000,000**

Ten Millions	Millions	Hundred Thousands	Ten Thousands	Thousands	Hundreds	Tens	Ones
1	0	0	0	0	0	0	0

Place value chart for one hundred million

Greatest eight digit number is **99,999,999**, we read it as “Ninety nine million nine hundred ninety and nine thousand nine hundred and ninety nine”.

When we add 1 to 99,999,999, we get **100,000,000**
Read as “One Hundred Million”
i.e. $99,999,999 + 1 = 100,000,000$

Unit 1 NUMBERS AND ARITHMETIC OPERATIONS (Numbers)

The place value chart for 100,000,000

Hundred Millions	Ten Millions	Millions	Hundred Thousands	Ten Thousands	Thousands	Hundreds	Tens	Ones
1	0	0	0	0	0	0	0	0

Representation on the abacus will be



Example 1:

Identify the place value and value of coloured digit in 2,546,789

M	H-Th	T-Th	Th	H	T	O
2	5	4	6	7	8	9

Digit 2 is million place; its place value is 2,000,000.

Example 2:

Write place value and value of each digit in 37,219,854

T-M	M	H-Th	T-Th	Th	H	T	O
3	7	2	1	9	8	5	4

Digit	Place Value	Value
3	Ten million	30,000,000
7	Million	7,000,000
2	Hundred thousand	200,000
1	Ten thousand	10,000
9	Thousand	9,000
8	Hundred	800
5	Ten	50
4	ones	4

Teacher's Note

Teacher should teach the concept of place value by using abacus on the blackboard.

EXERCISE 1.2

1 Identify the place value and value of the coloured digit.

(i) 1 **2**,345,678

(ii) 5 **8**,923,107

(iii) **1**00,000,000

(iv) 23, **9**64,579

(v) 2 **5**,960, **2**38

(vi) **9**,62 **1**,382

(vii) 8, **9**9 **8**,776

(viii) **7** **6**,905,851

2 Write down the value of every digit in the following numbers.

(i) 9,234,513

(ii) 50,120,306

(iii) 3,567,899

(iv) 36,564, 396

Read and write numbers up to one hundred million

In international system of units a number is split up into groups or periods. Each period consists of three digits.

The chart of periods and place values.

Millions			Thousands			Ones		
Hundred Millions	Ten Millions	Millions	Hundred Thousands	Ten Thousands	Thousands	Hundreds	Tens	Ones

Note: In reading a number, all the digits in the same period are read together along with its period (except the ones).

Commas are placed to separate the periods.

Reading and writing of a number “24,567,189”

Millions	Thousands	Ones
24	567	189
24	567	189

In words:



Twenty four **Million**, Five Hundred Sixty Seven **Thousand** and one hundred eighty nine

Unit 1 NUMBERS AND ARITHMETIC OPERATIONS (Numbers)

Example 1: Read and then write the given numbers in words.

- (i) 3,671,289 (ii) 45,612,378 (iii) 10,202,000

Solution:

(i) 3,671,289	Three million, six hundred seventy one thousand and two hundred eighty nine.
(ii) 45,612,378	Forty five million, six hundred twelve thousand and three hundred seventy eight.
(iii) 102,002,000	One hundred two million and two thousand.

Example 2: Write numbers from “one million and one” to “one million and fifteen” in figures.

Solution:

1,000,001, 1,000,002, 1,000,003, 1,000,004, 1,000,005,
1,000,006, 1,000,007, 1,000,008, 1,000,009, 1,000,010,
1,000,011, 1,000,012, 1,000,013, 1,000,014, 1,000,015.

Recognize numbers in words up to one hundred million

Nine digit numbers represent hundred millions such as 700,000,000, 850,000,000 and 710,999,999 read as “seven hundred million”, “eight hundred and fifty million” and “seven hundred ten million, nine hundred ninety nine thousand and nine hundred ninety nine” respectively.

100,000,000 is read as (One Hundred Million), it is the smallest 9-digit number.

Example 1: Write 53816432 in expanded form.

50,000,000 + 3,000,000 + 800,000 + 10,000 + 6000 + 400 + 30 + 2

or 50 millions + 3 millions + 8 hundred thousands
+ 10 thousands + 6 thousands + 4 hundreds + 3 tens + 2 ones

We read it as fifty three million, eight hundred sixteen thousand and four hundred thirty two.

Unit 1 NUMBERS AND ARITHMETIC OPERATIONS (Numbers)

Example 2: Separate numbers in millions, ten millions, and hundred millions from the following.

21,045,678, 100,000,000, 4,234,566, 2,005,127
and 55,566,677

Solution:

Millions	Ten Millions	Hundred Millions
4,234,566	21,045,678	100,000,000
2,005,127	55,566,677	—

EXERCISE 1.3

1 Read and then write the following numbers in words.

- (i) 241,935 (ii) 4,312,687 (iii) 5,000,000
(iv) 25,134,564 (v) 100,000,000 (vi) 9,264,387
(vii) 50,001,000 (viii) 4,109,200 (ix) 99,990,090

2 Write the following numbers in figures.

- (i) Two million, ninety thousand and sixty eight.
(ii) Thirty million, six hundred thousand and forty five.
(iii) One hundred million.
(iv) Twenty million and twenty.
(v) Ninety million and sixty seven thousand.

3 Write numbers from two million to two million twenty in figures.

4 Write the missing numbers.

- (i) 2,450,761, 2,451,761, 2,452,761, _____, _____, _____,
(ii) 7,000,300, 7,000,400, 7,000,500, _____, _____, _____,
(iii) 67,213,415, 67,223,415, 67,233,415, _____, _____, _____,

5 Separate numbers in millions, ten millions and hundred millions from the following.

2,456,178, 22,233,341, 1,000,000, 10,000,000,
100,000,000, 2,561,000 and 20,001,010

Unit 1 NUMBERS AND ARITHMETIC OPERATIONS (Numbers)

Compare and order numbers up to 8-digits.

We already know the rules of comparing numbers.
Let us recall them

Rule 1: When we compare two numbers, the number with less number of digits is always less and a number with more digits is always greater.

Example: Compare the following numbers.

- (i) 3,456,712 and 92,315,612
- (ii) 60,123,000 and 9,999,999

Solution:

- (i) $3,456,712 < 92,315,612$ because 3,456,712 has less number of digits than 92, 315, 612
- (ii) $60,123,000 > 9,999,999$ because 60,123,000 has more number of digits than 9, 999, 999

Rule 2: Two numbers having same number of digits are equal if digits at corresponding positions are same.

Example: Compare 2,456,127 and 2,456,127

Solution: $2,456,127 = 2,456,127$

- Both numbers have same number of digits.
- Write each number as the corresponding place value.
- We find digits at each position are same.

So, $2,456,127 = 2,456,127$

Rule 3: If two numbers have same number of digits then we compare the different digits of the higher place values.

Example: Compare 24,513,105 and 24,367,999

Solution: Both are 8-digits numbers.

Process as given in the above example:

- First we compare digits of higher place value.

Teacher's Note

Teacher should ensure enough practice by using the rules of comparison of numbers.

Unit 1 NUMBERS AND ARITHMETIC OPERATIONS (Numbers)

- Here the number have same digit 24 at million place.
- So, we compare next digit at hundred thousand place.
- Here $5 > 3$ (Hundred thousands)

So, $24,513,105 > 24,367,999$

Arranging numbers in orders:

Example: Arrange 5,671,231, 341,267, 90,000,000, 5,767,237.
in ascending and descending orders.

Solution: Using the rules of comparison of numbers.

Ascending Order

341,267, 5,671,231, 5,767,237, 90,000,000

Descending Order

90,000,000, 5,767,237, 5,671,231, 341,267

EXERCISE 1.4

1 Compare the following numbers using symbols $<$, $>$ and $=$ respectively.

- (i) 52,001,000 and 345,912
- (ii) 2,456,123 and 24,345,611
- (iii) 7,123,400 and 8,567,001
- (iv) 92,333,444 and 92,315,617
- (v) 24,000,008 and 24,000,005

2 Write the following numbers in ascending and descending orders.

- (i) 3,174,215, 3,741,512, 3,076,005
- (ii) 95,123,415, 95,312,415, 95,113,417
- (iii) 59,178,215, 59,296,712, 52,111,222, 58,110,176
- (iv) 14,111,920, 14,160,000, 13,200,415, 13,100,219

1.2 ADDITION

Add numbers up to 6-digits

We have learnt in class III to add numbers up to 4 digits.

Example: Add 8420 and 3910. **Example:** Add 45093 and 3421.

Solution:

$$\begin{array}{r} \textcircled{1} \\ 8420 \\ + 3910 \\ \hline \text{Sum} \quad 12330 \end{array}$$

So, $8420 + 3910 = 12,330$

Solution:

$$\begin{array}{r} \textcircled{1} \\ 45093 \\ + 3421 \\ \hline \text{Sum} \quad 48514 \end{array}$$

So, $45,093 + 3421 = 48,514$

Let us apply the addition rules to add numbers up to 6 digits.

Example 1: Add 85765 and 37071

Solution:

	T-Th	Th	H	T	O	
	①		①			
	8	5	7	6	5	
+	3	7	0	7	1	
	12	2	8	3	6	

Step1: Add ones
 $5 + 1 = 6$ ones

Step2: Add tens
 $6 + 7 = 13$ tens
 Write 3 below tens and Carry 1 to hundred column

Step3: Add hundreds
 $1 + 7 + 0 = 8$ hundreds

Step4: Add thousands
 $5 + 7 = 12$ thousand,
 Write 2 below thousand and regroup 10 thousand to one ten thousand

Step5: Add ten thousands
 $1 + 8 + 3 = 12$ T-Th
 Write 2 below T-Th and regroup 10 ten thousands to 1 hundred thousand

Hence $85,765 + 37,071 = 122,836$

Teacher's Note

Teacher should give some more examples to the students for practice by changing digits at place values.

Unit 1 NUMBERS AND ARITHMETIC OPERATIONS (Addition)

Example 2: Add 348,754 and 343,445

Solution:

$$\begin{array}{r} \textcircled{1}\textcircled{1} \\ 348754 \\ + 343445 \\ \hline 692199 \end{array}$$

So, $348,754 + 343,445 = 692,199$

EXERCISE 1.5

1. Solve the following.

(i)

$$\begin{array}{r} 25431 \\ + 41245 \\ \hline \\ \hline \end{array}$$

(ii)

$$\begin{array}{r} 92345 \\ + 50178 \\ \hline \\ \hline \end{array}$$

(iii)

$$\begin{array}{r} 107236 \\ + 298017 \\ \hline \\ \hline \end{array}$$

(iv)

$$\begin{array}{r} 349629 \\ + 201548 \\ \hline \\ \hline \end{array}$$

(v)

$$\begin{array}{r} 964328 \\ + 428961 \\ \hline \\ \hline \end{array}$$

(vi)

$$\begin{array}{r} 999555 \\ + 312016 \\ \hline \\ \hline \end{array}$$

2. Add the following.

- | | |
|---------------------------|----------------------------|
| (i) 24,317 and 90,235 | (ii) 67,999 and 249,982 |
| (iii) 392,612 and 722,334 | (iv) 591,023 and 942,589 |
| (v) 217,640 and 921,079 | (vi) 555,566 and 778,896 |
| (vii) 500,983 and 645,008 | (viii) 910,052 and 881,223 |

Unit 1 NUMBERS AND ARITHMETIC OPERATIONS (Addition)

Solve real life problems involving addition of numbers up to 6-digits

We do the addition of numbers up to 6-digits in our daily life in routine.

Example: There are **321,876** female and **313,589** male in a town. What is the total number of people in the town.

Solution: Number of female = **321876**
Number of male = **313589**

Total number of male and female in the town = **635,465**

$$\begin{array}{r} 321876 \\ + 313589 \\ \hline \text{Sum } 635465 \end{array}$$

EXERCISE 1.6

1. Government of Sindh spent Rs 581,034 on construction of one road and Rs 347,083 on another. Find the total amount spent in construction of both roads?
2. In an examination, 27,514 girls students and 20,328 boys students were appeared. What is the total number of students appeared in the examination?
3. Bismah purchased computers for Rs 857,600 and Aleesha purchased for Rs 641,200. What is the total amount they spent for purchasing computers?
4. In a two cricket match series Pakistan vs India, 55,384 people came to watch the first match at National Stadium Karachi and 43,298 people came for the second match in the Qaddafi Stadium Lahore. How many people came to watch the series in total?
5. A company manufactures 865,271 bicycles and another company manufactures 725,059 bicycles in a year. What is the total production of both companies in a year?
6. Pakistan Railways carried 347,180 kg of mangoes in first week and 449,130 kg mangoes in next week. How many kilograms of mangoes were carried altogether?

1.3 SUBTRACTION

Subtract numbers up to 6-digits

We have learnt in class III to subtract numbers up to 4 digits.

Example 1: Subtract 1,374 from 2,481

Solution:

$$\begin{array}{r} 2481 \\ - 1374 \\ \hline \end{array}$$

Difference 1107

Hence,
 $2,481 - 1,374 = 1,107$

Example 2: Solve $51,432 - 40,028$

Solution:

$$\begin{array}{r} 51432 \\ - 40028 \\ \hline \end{array}$$

Difference 11404

Hence,
 $51,432 - 40,028 = 11,404$

Example 3: Subtract 45,912 from 85,145

Solution:

	T-Th	Th	H	T	O
	8	5	1	4	5
	7	14	11		
	8	5	1	4	5
	4	5	9	1	2
	3	9	2	3	3

Step 1: Subtract ones

$5 - 2 = 3$ ones

Step 2: Subtract tens

$4 - 1 = 3$ tens

Step 3: Subtract Hundreds

$1 - 9$ H is not possible
 make 1 H to 11 H borrowing 1 from thousands. Now $11 - 9 = 2$ H.

Step 4: Subtract Thousands.

After giving 1Th, 5Th becomes 4Th
 So, $4\text{Th} - 5\text{Th}$ is not possible. Make 4Th to 14Th by borrowing 1 from Ten Thousands. Now $14\text{Th} - 5\text{Th} = 9\text{Th}$.

Step 5: Subtract Ten Thousands

$7\text{ TTh} - 4\text{ TTh} = 3\text{ TTh}$

Hence $85,145 - 45,912 = 39,233$

Teacher's Note

Teacher should solve some more examples using regrouping in order to clear the concept of subtraction.

Unit 1 NUMBERS AND ARITHMETIC OPERATIONS (Subtraction)

Example 4: Subtract 438,905 from 557,942

Solution:

$$\begin{array}{r} \overset{(4)}{5}\overset{(17)}{5}\overset{(3)}{7}\overset{(12)}{9}42 \\ - 438905 \\ \hline 119037 \end{array}$$

$$557,942 - 438,905 = 119,037$$

EXERCISE 1.7

1 Solve the following.

(i)
$$\begin{array}{r} 32164 \\ - 20053 \\ \hline \end{array}$$

(ii)
$$\begin{array}{r} 583729 \\ - 21678 \\ \hline \end{array}$$

(iii)
$$\begin{array}{r} 627948 \\ - 16328 \\ \hline \end{array}$$

(iv)
$$\begin{array}{r} 793854 \\ - 185035 \\ \hline \end{array}$$

(v)
$$\begin{array}{r} 790000 \\ - 628453 \\ \hline \end{array}$$

(vi)
$$\begin{array}{r} 894093 \\ - 847141 \\ \hline \end{array}$$

2 Subtract the following:

(i) 43,210 from 98,765

(ii) 39,072 from 273,194

(iii) 99,999 from 100,000

(iv) 537,864 from 700,000

(v) 685,439 from 874,189

(vi) 721,059 from 751,342

(vii) 781,500 from 871,600

(viii) 894,354 from 994,354

(ix) 990,001 from 991,000

(x) 184,019 from 765,129

(xi) 853,492 from 89,1400

(xii) 493,994 from 943,002

Unit 1 NUMBERS AND ARITHMETIC OPERATIONS (Subtraction)

Solve real life problems involving subtraction of numbers up to 6-digits

We use subtraction of numbers in our daily life in routine. Let us understand the process by following example.

Example: There are **847,385** chicken in a farm. From them **312,793** were sold. Find the number of chicken remain in the farm.

Solution: Number of chicken = 847385

Number of chicken sold = 312793

So, there are **534,592** chicken in the farm.

Difference

$$\begin{array}{r} 847385 \\ - 312793 \\ \hline 534592 \end{array}$$

EXERCISE 1.8

- 1 Government spent Rs 985,000 on renovation of two schools. If Rs 539,450 spent on one school, find the amount spent on the other school.
- 2 In an annual examination of grade IV 57,986 students appeared, from which 43,985 students passed. How many students did not pass the examination?
- 3 Aslam and Atif invested Rs 658,700 in establishing cattle farm. Share of Aslam is Rs 385,780. What is the share of Atif?
- 4 Profit of a trading of Cotton Company in two months is Rs 320,000. If Rs 139,998 is profit in one month, find the profit in the other month?
- 5 On the first day of eid 3,955 people visited the zoo and on the second day 3,843 people visited. How many more people visited the zoo on first day of Eid?
- 6 A poultry farm had 89,534 hens from which 43,294 hens died due to bird flu. Find the number of hens left in the poultry farm.

1.4 MULTIPLICATION

Multiply numbers up to 5-digits by numbers up to 3-digits

We have learnt in class III to multiply 2-digit numbers by 1-digit number. As we know that the process of multiplication is repeated Addition.

Let us learn the process by following examples.

Example 1: Multiply 24 by 3.

Solution:

$$\begin{array}{r} 24 \\ \times 3 \\ \hline 72 \end{array}$$

Product = 72 Hence $24 \times 3 = 72$

Example 2: Multiply 34251 by 32.

Solution:

$$\begin{array}{r} \textcircled{1} \quad \textcircled{1} \\ 34251 \\ \times 32 \\ \hline 68502 \\ + 102753 \times \\ \hline 1096032 \end{array}$$

Steps 1:

Multiply by 2 and write the numbers as their place value.

Steps 2:

Put a cross on the ones place which indicate that we multiply the number with 3-tens.

Steps 3:

(By Adding)

So, $34251 \times 32 = 1,096,032$

Example 3:

Multiply 40,329 by 123

Solution:

$$\begin{array}{r} \textcircled{1} \\ 40329 \\ \times 123 \\ \hline \textcircled{1} \textcircled{2} \textcircled{1} \\ 120987 \\ 80658 \times \\ + 40329 \times \times \\ \hline 4960467 \end{array}$$

So, $40329 \times 123 = 4960467$

Example 4:

Multiply 23,415 by 382

$$\begin{array}{r} \textcircled{1} \textcircled{1} \textcircled{1} \\ 23415 \\ \times 382 \\ \hline \textcircled{1} \textcircled{1} \textcircled{1} \\ 46830 \\ 187320 \times \\ + 70245 \times \times \\ \hline 8944530 \end{array}$$

So, $23415 \times 382 = 8944530$

EXERCISE 1.9

1 Solve the following:

(i) 1632×23

(iii) 6314×52

(v) 51389×562

(vii) 65123×316

(ii) 2341×70

(iv) 2109×84

(vi) 43851×725

(viii) 74156×163

2 Multiply:

(i) $11689 \text{ by } 100$

(iii) $25701 \text{ by } 553$

(v) $41078 \text{ by } 203$

(vii) $54321 \text{ by } 321$

(ii) $21499 \text{ by } 120$

(iv) $32145 \text{ by } 152$

(vi) $12345 \text{ by } 123$

(viii) $89713 \text{ by } 401$

Solve real life problems involving multiplication

We do the multiplication of numbers in our daily life in routine. Let us understand the process by following example.

Example: A factory produce **28543** toffees in one shift. Find the number of toffees in **132** such shifts.

Solution:

$$\begin{array}{r}
 \textcircled{2} \textcircled{1} \textcircled{1} \\
 28543 \\
 \times 132 \\
 \hline
 57086 \\
 85629 \times \\
 + 28543 \times \times \\
 \hline
 3767676
 \end{array}$$

Toffees produced in one shift
Number of shifts

So,
Total number of toffees are **3,767,676**

EXERCISE 1.10

- 1 There were 5,324 bottles of cold drinks loaded in a truck. How many bottles will be loaded in 132 trucks?
- 2 There are 630 students in a school. Each student pays Rs 1,200 as monthly fees. How much fee is collected by the school in a month?
- 3 Price of a motorcycle is Rs 35,800. Find the price of such 325 motorcycles.
- 4 A poultry farm produced 43,290 eggs in a day. How many eggs will be produced in 400 days?
- 5 A town uses 45,038 liters of water in a week. How much water will be used in 890 weeks?
- 6 A family spent Rs 15,980 to purchase food items for a month. What amount will be spent by 580 such families?

1.5 DIVISION

Divide numbers up to 4-digits by numbers up to 2-digits

We have learnt in class III to divide 2-digit numbers by 1-digit numbers. We use division in daily life. It is repeated subtraction.

Example 1:

Divide 56 by 8

$$\begin{array}{r}
 \text{Quotient} \\
 \text{Divisor } 8 \overline{) 56} \\
 \underline{- 56} \\
 \text{Remainder } 0
 \end{array}$$

Hence, $56 \div 8 = 7$

Example 2:

Divide 561 by 11

$$\begin{array}{r}
 \text{Quotient} \\
 11 \overline{) 561} \\
 \underline{- 55} \\
 11 \\
 \underline{- 11} \\
 0
 \end{array}$$

Hence, $561 \div 11 = 51$

Teacher's Note

Teacher should teach the division of numbers by the concept of repeated subtraction.

Example 3: Divide 975 by 15

Solution:

$$\begin{array}{r}
 65 \\
 15 \overline{) 975} \\
 \underline{- 90} \\
 75 \\
 \underline{- 75} \\
 00
 \end{array}$$

Hence,

$$975 \div 15 = 65$$

Explanations

- We start division from the highest value digit.
- If it is less than the divisor, we join the next digit.
- So we get 97.
- We have to divide 97 by 15, so we count maximum of multiples of 15 which can be subtracted from 97 which is 6 because $15 \times 6 = 90$. If we take 7 multiples then it will be $15 \times 7 = 105$ which is greater than 97. Write 90 below 97. Subtract 90 from 97 as shown.
- We write 6 as quotient.
- Take down the next digit which is 5 and we get 75. Now again repeat the process.

EXERCISE 1.11

1 Solve the following:

(i) $6744 \div 12$

(ii) $3795 \div 15$

(iii) $7293 \div 13$

(iv) $9384 \div 12$

(v) $2214 \div 18$

(vi) $9944 \div 22$

2 Divide the following numbers and find quotient.

(i) 4368 by 28

(ii) 8890 by 35

(iii) 5056 by 32

(iv) 6300 by 25

(v) 3920 by 16

(vi) 6642 by 18

3 Find the quotient and remainder when divisor is 35 and dividend is 5,075.

4 What will be the quotient and remainder for 5,696 as dividend and 16 as divisor.

5 If the divisor is 12 and dividend is 31,035. Find the quotient and remainder.

6 Find the quotient and remainder if 9,267 is divided by 15.

Solve real life problems involving division

We use the division of numbers in our daily life. Let us understand the process by the following example.

Example: 12 boxes of equal size contain **6816** toffees.
How many toffees are there in each box?

Solution:

Number of toffees = **6816**

Number of boxes = **12**

Number of toffees in each box = ?

Hence there are **568** toffees in each box.

Division:

$$\begin{array}{r} \text{568} \\ 12 \overline{) 6816} \\ \underline{-60} \\ 81 \\ \underline{-72} \\ 96 \\ \underline{-96} \\ 00 \end{array}$$

EXERCISE 1.12

- 1** A cloth is 6,272 metres long. If one suit takes 7 metres of cloth. How many suits can be made?
- 2** Monthly salary of 12 workers is Rs 7,032. What will be the salary of one worker if they are taking same salary?
- 3** Asif spent Rs 2,925 to buy 13 shirts of same price. What is the price of each shirt?
- 4** The weight of 16 sacks of flour is 4,496 kg. What is the weight of each sack?
- 5** Price of 25 cricket bats is Rs 3,075. What is the price of one bat?
- 6** One crate of bottles has capacity to put 36 bottles. How many crates required for 5,616 bottle.

1.6 USE MIXED OPERATIONS OF ADDITION & SUBTRACTION AND MULTIPLICATION & DIVISION

Apply fundamental operations of addition, subtraction, multiplication and division

Combined Operations must be done in this order

1. Deal with brackets
2. If $+/-$ and $\times/\div$ are not combined, then perform the operations from left to right in expression/equation
3. When $+/-$ and $\times/\div$ are combined, then first perform division and multiplication before addition and subtraction in expression/equation

Example 1: Solve: $86 - 34 + 62$

Solution: As both addition and subtraction operations are involved in the question, we have to perform both operations one by one or simply work from left to right

$$\begin{aligned} & 86 - 34 + 62 \\ = & \boxed{86 - 34} + 62 \\ = & \boxed{52 + 62} = 114 \end{aligned}$$

Example 2: Solve: $4 + 6 \div 2 \times 3$

Solution:

$$\begin{aligned} & = 4 + \boxed{6 \div 2} \times 3 \quad (\text{Perform division first}) \\ & = 4 + \boxed{3 \times 3} \quad (\text{Perform multiplication}) \\ & = 4 + 9 = 13 \end{aligned}$$

Example 3: Solve: $3 \times 4 + 2$

Solution: As multiplication and addition both are involved in question, we have to perform multiplication first then addition.

$$\begin{aligned} 3 \times 4 + 2 & = \boxed{3 \times 4} + 2 \\ & = 12 + 2 \\ & = 14 \end{aligned}$$